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CFD and Thermal Coupled Modeling for Cutter Cooling and Bit Optimization in Hard Drilling Applications

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Abstract

Efficient cutter cooling and thermal management are essential for enhancing Polycrystalline Diamond Compact (PDC) bit performance, extending drilling intervals and reducing bit trips. By accurately simulating bit hydraulics and cutter temperature profiles, engineers can address critical drilling challenges such as excessive cutter wear and reduced rate of penetration (ROP). This study presents a coupled Computational Fluid Dynamics (CFD) and thermal analysis approach for optimizing cutter cooling. The optimized bit design was successfully implemented in the field, achieving over a 200% increase in drilling distance and demonstrating a significant improvement in cutter wear condition.

In the coupled CFD-thermal model, the PDC diamond table, substrate, and bit body are represented with their respective thermal material properties. A PDC wear drilling model is integrated into the CFD framework at a prescribed bit dull condition to compute cutting forces and frictional heat across cutter wear flats. The heat flux is then applied within the CFD-thermal model to predict cutter temperatures under fluid cooling conditions. The influence of fluid velocity, fluid properties, and drilling speed on thermal behavior is evaluated. Nozzle placement, orientation, and junk slot geometry are optimized through multiple iterative design cycles guided by CFD-thermal models to maximum cutter cooling efficiency. The final bit design is field-tested in a hard and abrasive drilling application.

The coupled CFD and thermal modeling workflow was employed to evaluate the impact of drilling parameters and bit hydraulics design on cutter cooling and temperature. The analysis showed nozzle positioning and junk slot geometry are critical for achieving optimal near-cutter fluid velocity and effective cutter cooling. Guided by the cutter cooling model, multiple design iterations were conducted to optimize nozzle positions, orientations, and junk slot configurations, enhancing cutter cooling efficiency. Following thermal optimization, the cutter fluid velocity and corresponding temperatures were reduced effectively. This study also identified a nonlinear relationship between flow rate and cutter temperature reduction, indicating the existence of an optimal fluid velocity threshold beyond which cooling efficiency diminishes. The thermally optimized bit was field-tested in a challenging hard and abrasive drilling application. Field results demonstrated a more than 200% increase in drilling footage. The bit exhibited a significantly improved dull condition with smooth, fracture-free wear scars.

This study introduces a coupled CFD-thermal modeling workflow for optimizing bit hydraulics for cutter temperature and wear. Both simulation and field-testing results highlight the critical role of nozzle positioning and junk slot geometry in enhancing bit performance, particularly in hard and abrasive drilling environments. The findings demonstrate that the CFD-thermal model is a powerful and essential tool for cutter cooling optimization and should be considered a key component in the bit design process.

Introduction

PDC drill bits have been widely used for high-performance drilling in a wide range of formations due to their high ROP and ability to maintain cutting efficiency over extended intervals. However, a key challenge in optimizing PDC bit performance is managing thermal damage and abrasive wear on the cutters. Excessive heat at the cutter-rock interface can degrade the diamond layer, accelerate wear, and lead to premature bit failure, resulting in early Pull Out Of Hole (POOH). Effective cutter cooling is therefore essential to preserve bit integrity, maintain ROP, extend drilling intervals, and minimize non-productive time caused by bit trips.

Effective cooling and thermal management are critical to the performance of PDC cutters. Although PDC diamond is extremely hard and wear-resistant, it begins to degrade at elevated temperatures (G. D. Zhan 2022). While the diamond table has high thermal conductivity, the tungsten carbide substrate is less efficient at dissipating heat. This mismatch can result in localized heating during drilling, particularly in hard and abrasive formations, leading to graphitization, thermal cracking, and delamination of the cutters. Previous research has explored the mechanisms of heat generation and cooling in PDC bits (Watson 1997, D. a. Glowka 1986, D. a. Glowka 1986). While numerous studies have relied on empirical data and lab testing to correlate cutter wear with drilling parameters, CFD has emerged as a powerful tool for simulating the complex interactions between drilling fluid, bit geometry, and thermal behavior. For instance, the CFD study by Schnuriger et al (Schnuriger 2017) showed that curved nozzle geometry can improve the bit cleaning and cutter cooling efficiency. The research by Sion et al (Sion Llywelyn Roberts 2021) highlighted the importance of CFD modeling and showed bit hydraulics optimization using CFD can delay and migrate the thermal damage of PDC cutters.

Despite recent advancements, most CFD models have limitations that reduce their accuracy under realistic drilling conditions. Many models either assume a sharp, unworn PDC bit (Sion Llywelyn Roberts 2021) or treat fluid flow and cutter temperature as separate phenomena (Schnuriger 2017). In practice, cutter wear changes the contact area and frictional heat at the rock interface, directly influencing heat generation and temperature distribution (Detournay, Richard and Shepherd 2008, Glowka and Stone 1985, Zhan, et al. 2023). To address this gap, this paper introduces a methodology that combines a thermal-coupled CFD model with a PDC wear drilling model (Matthews III, Huang and Bomidi 2021, Huang, et al. 2025). The approach incorporates the thermal properties of the diamond table, tungsten carbide substrate, and bit body geometry. Different drilling models have previously been developed using both physics-based and data-driven methods (Pelfrene, et al. 2019, Xu, et al. 2024, Warren and Sinor 1986, Huang, Luu, et al. 2024, Matthews III, Huang and Bomidi 2021). In this study, a 3D drilling model is employed to simulate both sharp and worn cutter states to calculate cutting forces and frictional heat generation (Zhan, et al. 2021). This heat is then applied within the CFD-thermal model to predict cutter temperatures under various fluid cooling conditions. Key variables such as fluid velocity, fluid properties, and drilling speed are analyzed for their impact on thermal behavior. Additionally, the influence of bit design features, including nozzle placement, orientation, and junk slot geometry, on cooling efficiency is evaluated. Multiple design iterations, guided by CFD-thermal simulations, are performed to converge on an optimized bit configuration that maximizes cutter cooling and overall performance.

The final bit design was field-tested in a hard and abrasive drilling application, where effective thermal management is especially critical. The results showed a substantial improvement in performance: drilling footage increased by more than 200%, and post-run inspections revealed significantly lower cutter wear.

compared to the baseline design. These findings validate the effectiveness of the coupled modeling approach and demonstrate that integrating thermal-coupled CFD with wear modeling can significantly enhance bit performance. The study shows the importance of CFD-thermal analysis as an important tool for optimizing cutter cooling and should be considered a key component in the bit design process.

CFD and Thermal Coupled Model

Accurately predicting cutter temperatures in PDC bits requires a detailed understanding of both the heat generation at the cutter-rock interface and the cooling effects of drilling fluid. In this study, a thermal-coupled CFD framework was developed, integrating a validated 3D drilling mechanics model to simulate cutter behavior in both sharp and dull states (Matthews III, Huang and Bomidi 2021). As shown in Figure 1, the drilling model captures the geometry of wear scars on a prescribed dull bit across various depths of cut and calculates the corresponding drilling responses. Figure 1(a) illustrates the simulated contact geometry and the Weight On Bit (WOB) and aggressiveness at different ROP. Figure 1(b) compares the simulated ROP vs. WOB and ROP vs. Torque On Bit (TOB) relationships with full-scale downhole drilling test data, showing strong agreement in both sharp and dull conditions. Figure 1(c) further validates the model by demonstrating close alignment between simulated wear scar geometry and field-scanned cutter wear profiles. Using this validated drilling model, heat generated from cutter-rock interactions was calculated and applied as thermal boundary conditions in the CFD-thermal model. This integration enables precise prediction of cutter temperatures under various fluid cooling scenarios, providing a robust foundation for optimizing thermal management in bit design.

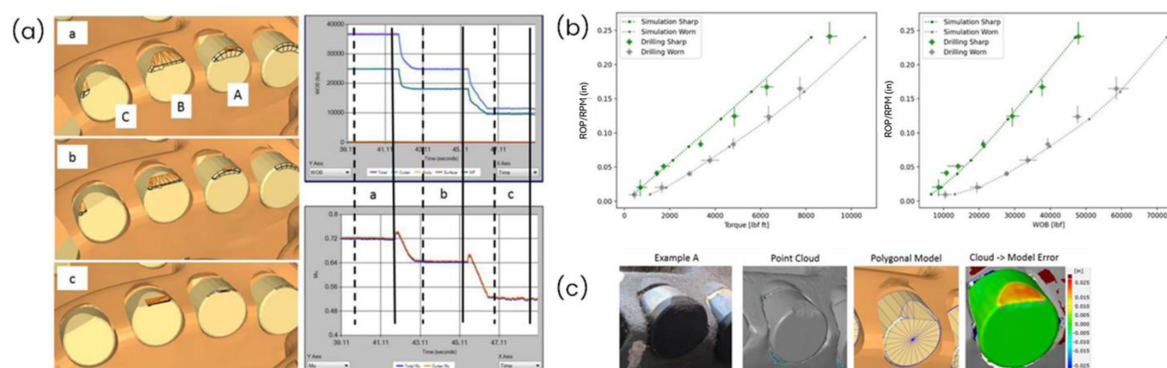


Figure 1—3D Drilling mechanics model (a) rock-cutter interaction and drilling responses at different DOCs (b) drilling responses of both sharp and dull state bits: simulation vs. lab testing (c) drilling model captures the wear scar geometry

Accurate modeling of cutter heat generation in both sharp and dull conditions is essential for evaluating the cooling efficiency of PDC bits. Cutter heat generation is directly proportional to the cutting power, which varies significantly depending on the wear state of the cutter. Figure 2 illustrates the normalized distribution of cutter cutting power for both sharp and dull bit conditions. In the sharp state, cutting power is primarily influenced by the tangential cutting force, the cutter's radial position, and the cutter rotational speed. However, as the bit becomes dull, the cutting power becomes largely governed by the worn cutter area rather than the tangential force. This transition leads to a notable redistribution of cutting work across the bit, with the highest cutting power shifting toward the shoulder region at a larger radial position, as shown in Figure 2. In addition to the distribution of heat generation, modeling results show that the cutter's peak temperature can increase by more than 300% when transitioning from a sharp to a dull state. Therefore, to accurately predict cutter temperatures and assess cooling performance, it is critical to account for heat generation in both sharp and dull cutter states.

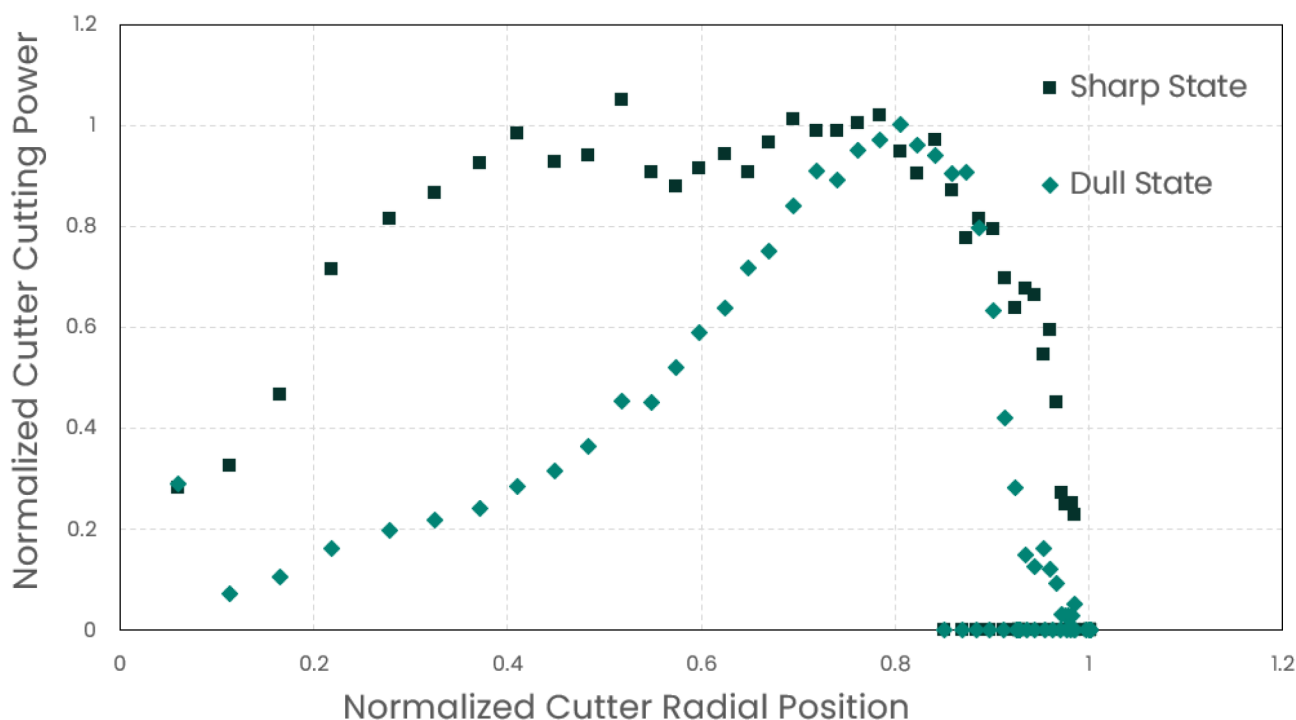


Figure 2—Normalized cutter work rate distributions at sharp and dull states

Figure 3 illustrates the integration of the bit drilling simulation model with the coupled CFD-thermal model to evaluate cutter temperature and cooling performance. As shown in Figure 3(a), the process begins with a bit drilling simulation, which provides detailed outputs such as cutter forces and frictional heat. These results are then used as thermal boundary conditions in the subsequent CFD analysis in Figure 3(b). In the thermal-coupled CFD model, both solid domains, including the bit body and cutter elements, and the fluid domain such as junk slot region are explicitly modeled and meshed. This is in contrast to a non-thermal CFD model, where only the fluid domain is typically required. The thermal properties of materials, such as specific heat capacity and density, are included in the simulation. The non-thermal CFD model solves only the momentum and continuity equations to capture fluid flow behavior, the thermal-coupled model also includes the energy equation and accounts for heat transfer across fluid-solid interfaces. Additionally, the CFD-thermal model incorporates the effects of bit rotation on fluid dynamics and cutter cooling, which are important for realistic thermal analysis.

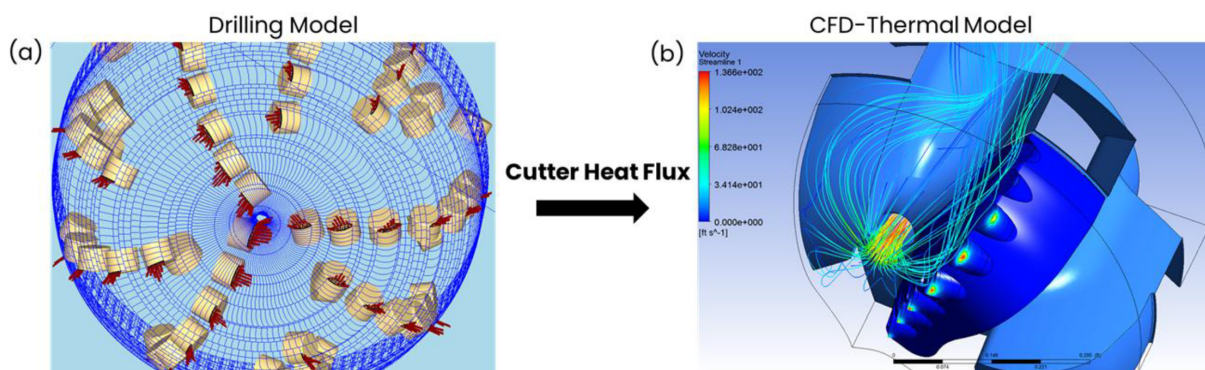


Figure 3—Integration of drilling response model with coupled CFD-thermal model for cutter temperature modeling

A steady-state CFD simulation is performed to evaluate the stabilized cutter temperature distribution. Figure 3(b) presents the simulated temperature field for a bit with a predefined dull condition. The results

indicate that the highest cutter temperature (highlighted in red) is located at the wear flat of the shoulder cutter, where thermal loading is most severe. The color of the streamlines in the figure represents fluid velocity, providing insight into the cooling flow behavior around the cutters.

To verify the CFD thermal-coupled model, full-scale atmospheric bit drilling tests were conducted (Ledgerwood and Kelly JR. 1991, Armin, et al. 2023). A novel measurement technique was employed to capture cutter tip temperatures during drilling. As illustrated in Figure 4(b), a narrow hole was precisely fabricated within the diamond table of the cutter to accommodate a thermocouple. This hole was carefully dimensioned to extend to the cutter tip, enabling direct temperature measurements at the cutting interface. The thermocouple-equipped cutter was placed at a selected location on the drill bit to monitor thermal behavior under realistic drilling conditions. The measured cutter temperatures from these tests are presented in Figure 4(a). Multiple test runs were performed under varying drilling parameters to assess the relationship between operational conditions and thermal response.

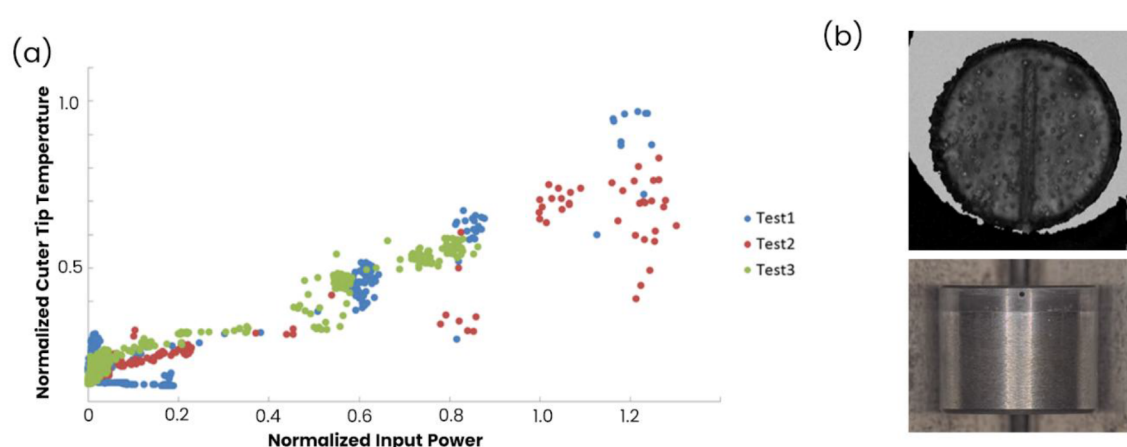


Figure 4—Cutter tip temperature measurement in full-bit rock drilling tests (a) normalized cutter tip temperature vs. normalized input power (b) PDC cutter with a fabricated hole inside the diamond table to accommodate thermal couple

Experimental results revealed a linear correlation between bit drilling power and cutter temperature, indicating that increased mechanical energy input directly contributes to elevated thermal loading at the cutter interface. The similar drilling parameters were applied to both the bit drilling model and the thermal-coupled CFD model. The CFD results successfully reproduced the observed temperature-power relationship, confirming the model's capability to predict cutter thermal behavior under realistic drilling conditions.

Modeling Results

The verified CFD thermal-coupled model is employed to investigate the influence of drilling parameters on cutter temperature. Figure 5 presents simulation results showing the relationship between normalized maximum cutter temperature and near-cutter fluid velocity. As expected, an increase in fluid velocity near the cutter enhances convective heat transfer, resulting in a reduction in cutter temperature. However, the analysis reveals a distinct velocity range. When the normalized fluid velocity is between 0.0 and 1.3, the increases in fluid velocity lead to a significantly more effective reduction in cutter temperature. Within this range, cutter temperature decreases by approximately 36%. Beyond this threshold, from 1.3 to 2.5, the rate of temperature reduction diminishes, with only an additional 9% decrease observed. This indicates diminishing returns in cooling efficiency at higher fluid velocities. Near-cutter fluid velocity can be influenced by several design and operational parameters, including flow rate, nozzle size, and nozzle positioning. However, these parameters are often constrained by bit design limitations and field operational requirements. Therefore,

optimizing the bit's hydraulic configuration within these constraints is critical to achieving effective cutter cooling.

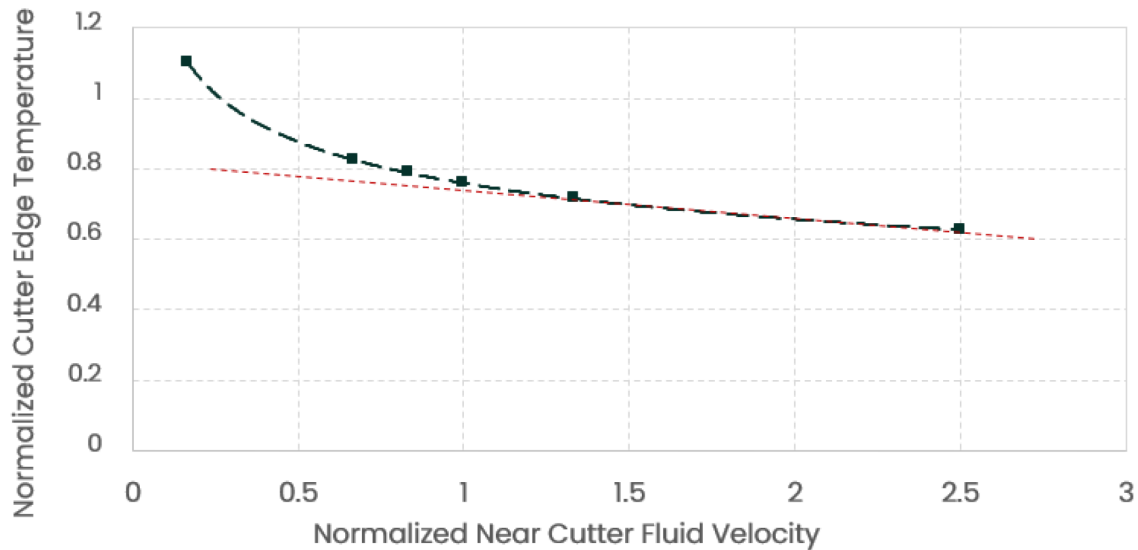


Figure 5—Simulated cutter temperature vs. near cutter fluid velocity

Figure 6 presents a comparison of the maximum cutter temperatures using water and diesel oil as drilling fluids. Water demonstrates a more efficient cooling capacity compared to diesel oil, primarily due to its higher specific heat capacity, thermal conductivity, and density. These thermal properties enable more efficient heat dissipation from the cutter interface. CFD simulation results indicate that the maximum cutter temperature during water-based drilling is approximately 10–20% lower than that observed with diesel oil. This highlights the significant influence of drilling fluid selection on cutter cooling effectiveness. Furthermore, the temperature difference between the two fluids is found to be dependent on the near-cutter fluid velocity. A more pronounced temperature difference is observed at higher fluid velocities.

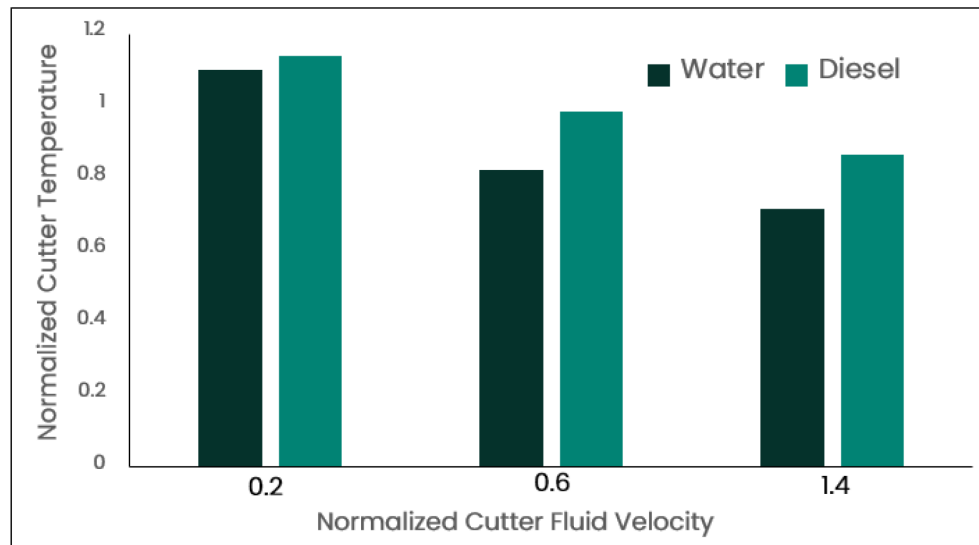


Figure 6—Cutter temperature at given cutter fluid velocity: water vs. diesel

To evaluate the impact of bit hydraulics design on cutter cooling performance, near-cutter fluid velocity was analyzed by varying nozzle location, nozzle tilt angle, and junk slot configuration relative to a baseline

bit design. Figure 7 compares the cutter fluid velocity distribution between the baseline nozzle position and a recessed nozzle configuration. In the recessed configuration, the nozzle is positioned farther from the rock surface, resulting in a greater standoff distance. Simulation results indicate that this recessed placement significantly reduces fluid velocity in the central cone and nose regions of the bit, specifically from Cutter 1 to Cutter 10. However, the influence of nozzle recession diminishes in the shoulder region. This result suggests that minimizing the standoff distance by positioning the nozzle closer to the rock surface enhances near-cutter fluid velocity and, consequently, improves cutter cooling efficiency and reduces cutter temperature. It is also important to note that the effectiveness of this configuration is influenced by the nozzle tilt angle. When the nozzle is tilted toward the shoulder region, the recessed position can have a more pronounced effect on cutter temperature in the shoulder area.

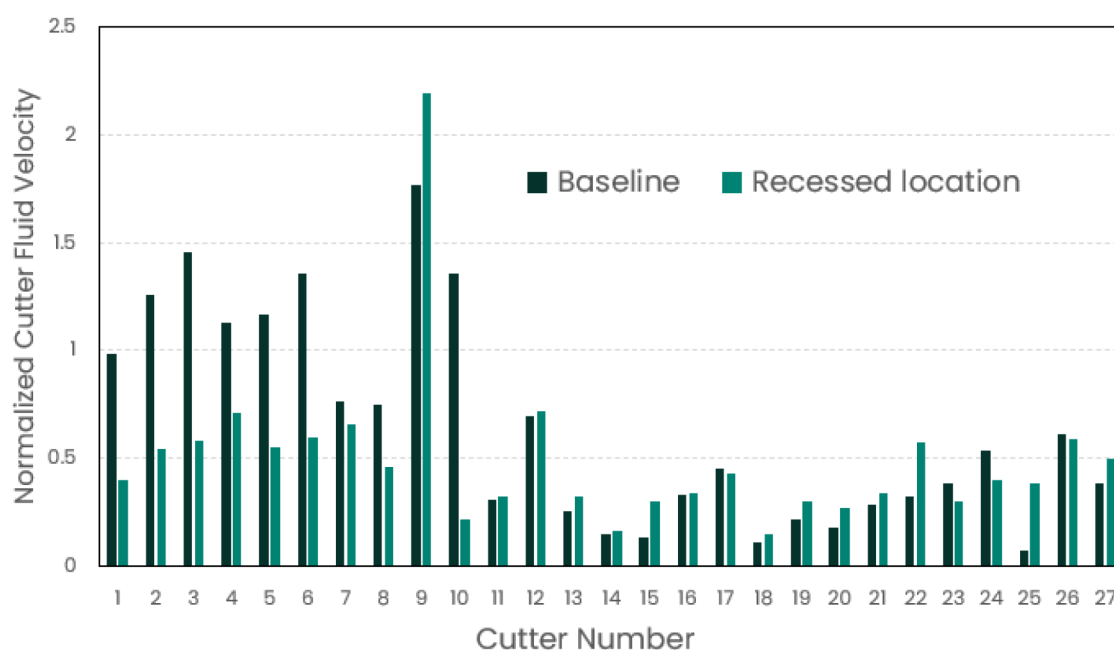


Figure 7—Cutter fluid velocity: baseline vs. recessed nozzle location

Figure 8 illustrates the impact of bit blade thickness on near cutter fluid velocity. Simulation results indicate that reducing blade thickness leads to a noticeable decrease of fluid velocity in the nose and shoulder regions of the bit. In these regions, 11 out of 15 cutters experience reduced fluid velocity when blade thickness is decreased, suggesting that increasing blade thickness can be an effective strategy for increasing fluid velocity and enhancing cutter cooling in these critical areas. However, while thicker blades may improve fluid delivery and cooling performance, they also introduce potential issues such as bit balling and hinder cuttings removal, which negatively affects drilling efficiency. As such, the application of thick blades is recommended for hard and abrasive formations with a low target ROP.

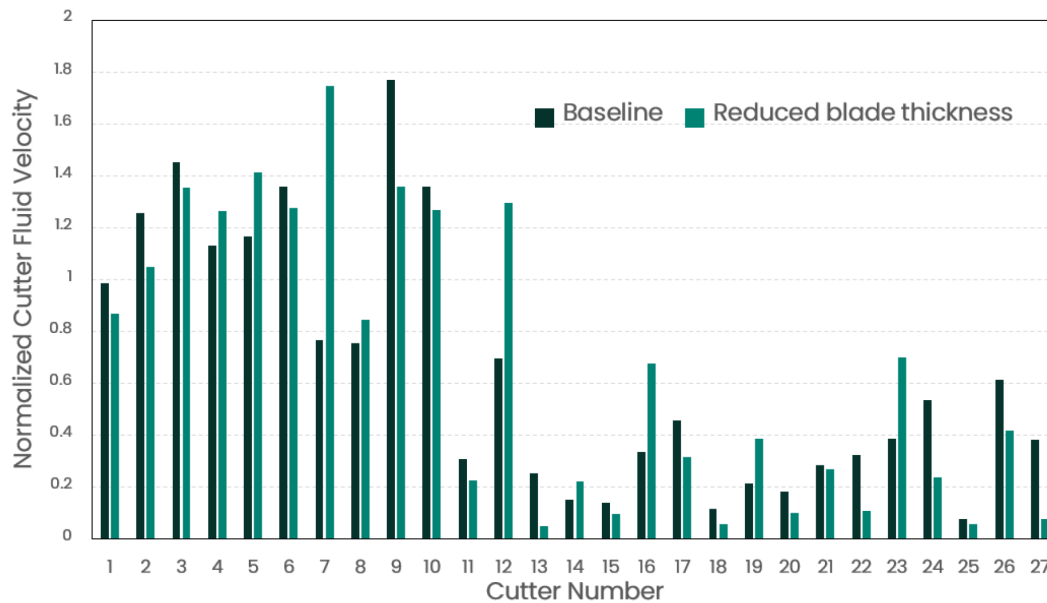


Figure 8—Cutter fluid velocity: baseline vs. reduced bit blade thickness

Figure 9 presents CFD simulation results illustrating the effect of nozzle tilt angle on near cutter fluid velocity. Three-bit configurations with varying nozzle tilt angles in the circumferential direction were analyzed to assess their influence. The results demonstrate that altering the nozzle tilt angle by ± 40 degrees significantly impacts fluid velocity in the cone, nose, and shoulder regions. For instance, at cutter 21, a change in nozzle tilt angle results in a fluid velocity variation exceeding 200%, highlighting the sensitivity of cutter cooling performance to nozzle orientation. This finding emphasizes the critical role of nozzle positioning and tilt angle in optimizing bit hydraulics. Careful design and adjustment of these parameters are essential to enhance fluid delivery to the cutter interface, improve cooling efficiency, and ultimately boost overall bit performance.

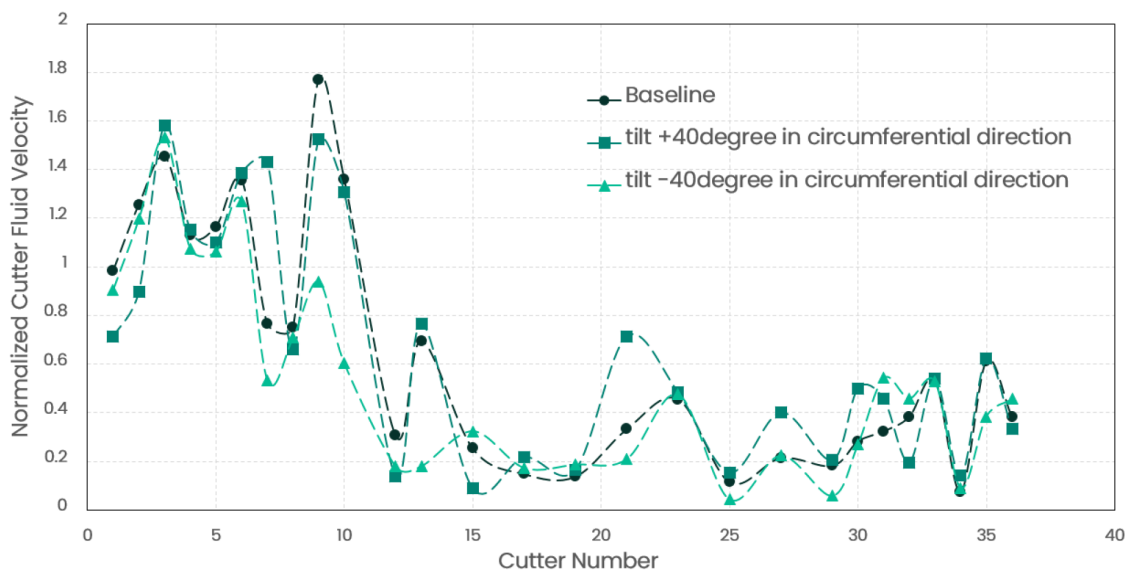


Figure 9—Cutter fluid velocity: baseline vs. nozzle tilt change

Bit Optimization and Field Application

The validated thermal-coupled CFD model was applied to optimize bit hydraulics and enhance cutter cooling performance for a hard and abrasive drilling application. The baseline bit design was a nine-blade PDC bit, which drilled less than 150 feet in an offset well before being POOH due to low ROP. Root cause analysis identified insufficient cutter cooling as a primary contributor to the performance issue. Specifically, the nozzle configuration in the nine-blade design did not provide adequate fluid coverage for all cutters across all blades. As shown in Figure 10(a), most cutters in the shoulder region exhibited normalized fluid velocities below 0.2, an indicator of inadequate convective cooling. To address this, the bit was redesigned with a reduced blade count from nine to seven blades, allowing for improved fluid distribution to each cutter. Additionally, the junk slot geometry and nozzle positions were optimized in iterations based on CFD modeling insights to enhance near cutter fluid velocity. The resulting fluid velocity distribution for the redesigned bit is shown in Figure 10(b), where most shoulder cutters now receive fluid velocities above the 0.2 threshold. CFD temperature simulations show that the enhanced cutter fluid velocity lowers peak cutter temperatures by over 20%. To further improve the cutter cooling, the StayCool PDC cutter geometry is also used in the new bit.

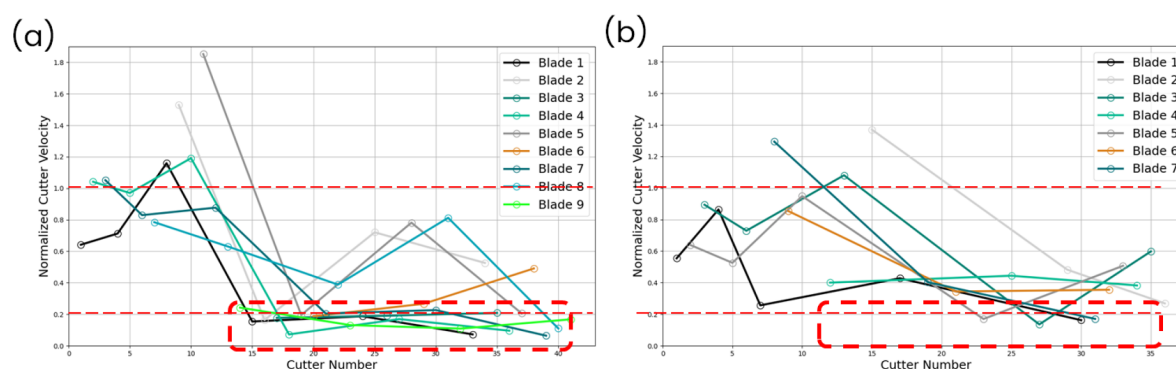


Figure 10—Cutter fluid velocity distribution: baseline bit (a) vs. hydraulics optimized bit (b)

The redesigned bit was field-tested in a comparable hard formation. It successfully drilled over 480 feet, representing a 200% increase in drilling footage compared to the baseline bit. Furthermore, the averaged ROP is more than doubled, demonstrating the effectiveness of CFD-guided hydraulics optimization in improving bit performance under challenging drilling conditions.

Figure 11(a) and 11(b) compare the post-run dull conditions of the baseline bit and the hydraulically optimized bit following field testing. The optimized bit, which incorporated CFD-guided hydraulics improvements, demonstrated significantly reduced cutter wear despite drilling a more than 200% footage. This improvement is attributed to enhanced cutter cooling, achieved through better fluid distribution, increased near cutter fluid velocity, and StayCool cutter geometry.

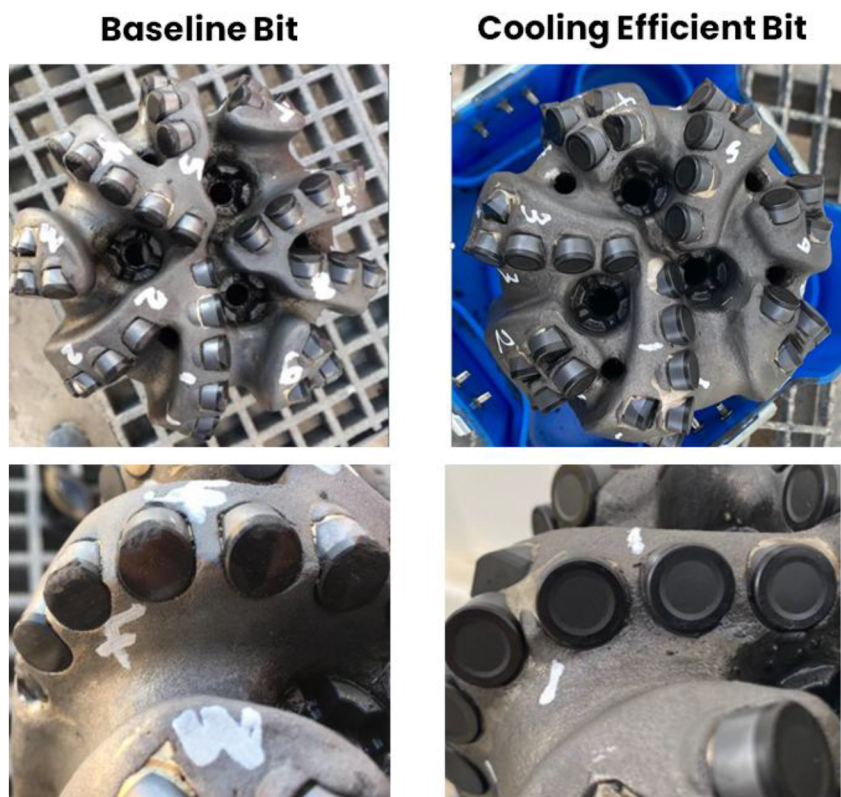


Figure 11—Field bit dull photos: baseline bit vs. cooling efficient bit.

Conclusions

In conclusion, this study presents an integrated modeling workflow that combines bit drilling model with a thermal-coupled CFD model to effectively address the challenges of cutter cooling and thermal management in PDC drill bits. Cutter's peak temperature can increase by more than 300% when transitioning from a sharp to a dull state. By incorporating realistic dull bit conditions and simulating frictional heat generation, the model enables accurate prediction of cutter temperatures across a range of drilling scenarios and bit designs. Through iterative optimization of nozzle placement, tilt angle, and junk slot geometry, guided by CFD simulation insights, the study demonstrates significant improvements in cutter cooling efficiency, achieving a reduction in peak cutter temperature more than 20%. Field validation in a hard and abrasive formation confirmed the model's effectiveness. The thermally optimized bit achieved over a 200% increase in drilling footage and exhibited substantially reduced cutter wear. The findings demonstrate that the CFD-thermal model is a powerful and essential tool for cutter cooling optimization and should be considered as a key component in the bit design process.

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